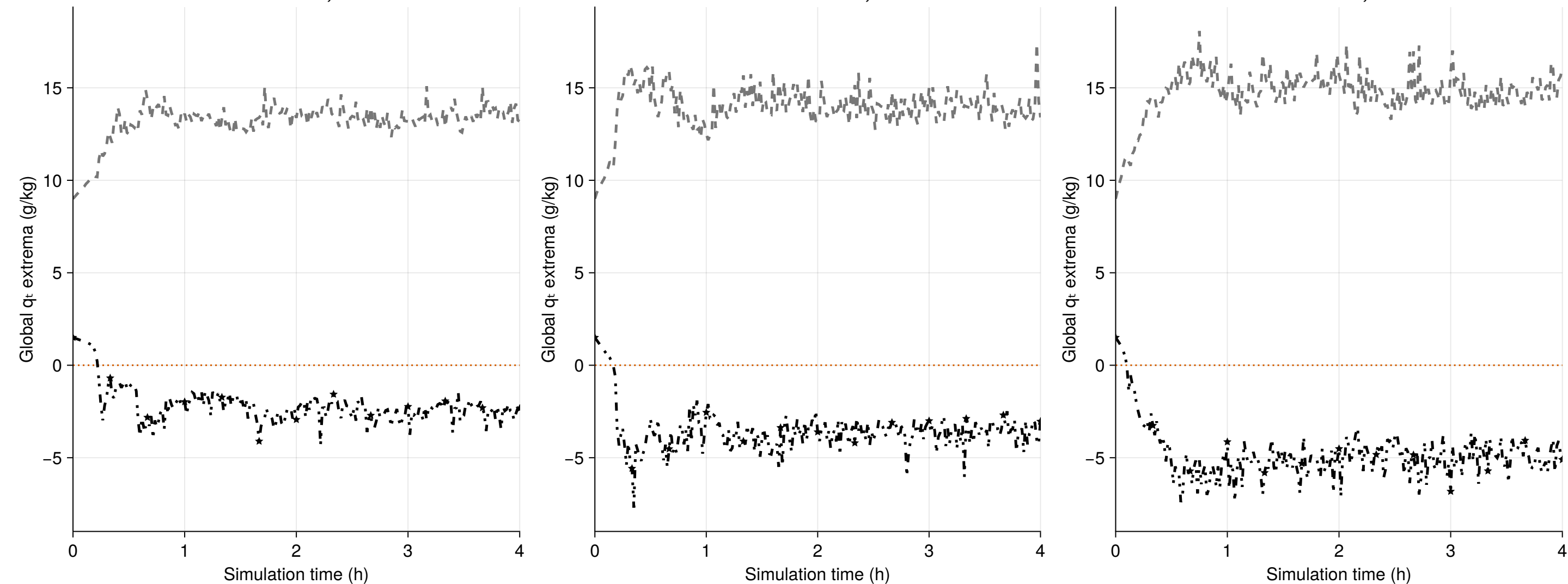


Unbounded Centered2 + Smagorinsky: moisture extrema

Coarse: $\Delta x = 80$ m, $\Delta z = 20$ m

Canonical: $\Delta x = 35$ m, $\Delta z = 5$ m

Fine: $\Delta x = 10$ m, $\Delta z = 5$ m



- · - Minimum q_t (star markers) - - - Maximum q_t ····· Zero total water

Global spatial extrema sampled every 60 s; excursions between samples are not bounded by these curves.
Negative total water is physically inadmissible. Finite output does not establish physical validity; no clipping or WENO fallback was applied.